2025-10-23 - Servidio (7)

Previous lecture: 2025-10-17 - Valentini (6)
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Introduction and program

We will study the applications of plasma theory.
Example:
On the solar surface appears the Coronal Mass ejections

The first system is stable and the structure evolves in a very stable evolution.
Then a dishomogeneity grows, with fine scale turbulence, here the unstability appears.
It is a quick dynamical evolution which creates an explosion and the emission of a CME.
This phenomenon is unpredictable and nonlinear.

Equilibrium -> Gradients -> Instability -> Turbulence

We will discuss about three main topics:

1. Quasi equilibrium structures
2. Instabilities
3. Nonlinear regime
   We have applications of this in cosmology, SN remnants, Solar physics, ecc

The suggested books are

- Krall & Trivelpiece, *Plasma physics* (1973)
- Davidson, *Methods in nonlinear plasma physics* (1972)
- D. Bismand, *Nonlinear Magnetohydrodynamics* (1993)

From the ensemble to Vlasov equation

We begin our discussion considering an ensemble of particles with a fixed number of N particles which have position x_i and velocities v_i.
We could study them in a Lagrangian form

$$\begin{cases} \dot{\bar{x}}_i = \bar{v}_i \\ \dot{\bar{v}}_i = \bar{a}_i \end{cases}$$

with unspecified acceleration $\bar{a}_i$.
We could aso use the Eulerian study in one spatial dimension and one velocity dimension (1d 1v system).
We will have a phase space at every time t.
We need to reduce the degree of freedom by disctretizing the 1d 1v dimensions (course grain of phase space)

We can choose an arbitrary Δx and Δv
In one cell $\Delta x \Delta v$ we have $\Delta N < \infty$ number of particles

$$\Delta N = \sum f(\bar{x}, \bar{v}, t)\, \Delta x \Delta v$$

with $f(\bar{x}, \bar{v}, t)$ being the probability (or velocity) distribution function.
The master equation is

$$\frac{d}{dt} f(\bar{x}(t), \bar{v}(t), t) = 0$$

$$\frac{\partial f}{\partial t} + \frac{\partial \bar{x}}{\partial t}\frac{\partial f}{\partial \bar{x}} + \frac{\partial \bar{v}}{\partial t}\frac{\partial f}{\partial \bar{v}} = 0$$

Which gets us to the Vlasov equation

$$\frac{\partial f}{\partial t} + \bar{v} \cdot \nabla f + \bar{a} \cdot \nabla_{\bar{v}} f = 0$$

The different moments of Vlasov equation

Zeroth moment

If we consider different species (with different index α), we will have

$$f_\alpha(\bar{x}, \bar{v}, t); \qquad n_\alpha(\bar{x}, t) = \int_{-\infty}^{\infty} f_\alpha(\bar{x}, \bar{v}, t)\, d^3v$$

with $n_\alpha(\bar{x}, t)$ being the density of particles (zeroth moment).
We also have the total number of particles in the system at a given time (which is adimensional)

$$N_\alpha(t) = \int_{-\infty}^{\infty} n_{\alpha(\bar{x},t)}\, d^3x$$

We can do a dimensional analisys on f

$$[f] = \frac{\#}{L^3 (L/T)^3} = \# \frac{T^3}{L^6}$$

First moments

Velocity

$$\bar{v}(\bar{x}, t) = \frac{\int \bar{v} f(\bar{x}, \bar{v}, t)\, d^3v}{\int f(\bar{x}, \bar{v}, t)\, d^3v} = \frac{\int \bar{v} f(\bar{x}, \bar{v}, t)\, d^3v}{n(\bar{x}, t)}$$

We note that the variable n at the denominator means that we are not allowed to work in vacuum (with $n(\bar{x}, t) = 0$). In that case we need some new physics.
We define the bulk velocity $\bar{U}$ as the mean velocities of the particles over a small region.
$\bar{U} = 0$ does not mean that particles are not moving.
In a closed bottle the air can reach a fraction of the light speed, but the bulk velocity will still be zero anyways.

The two fluids equations

We take the Vlasov-Mawell equations (for $\alpha = p, e \to f_\alpha(\bar{x}, \bar{v}, t)$)

$$\frac{\partial f_\alpha}{\partial t} + \bar{v} \cdot \nabla f_\alpha + \frac{q_\alpha}{m_\alpha}\left[\hat{E} + \bar{V} \times \bar{B}\right] \cdot \nabla_{\bar{v}} f_\alpha = 0$$

$$\begin{cases} \nabla \cdot \bar{E} = \dfrac{1}{\varepsilon_0} \sum_\alpha q_\alpha n_\alpha = \dfrac{1}{\varepsilon_0} \sum_\alpha \rho_{c,\alpha} \\ \nabla \times \bar{E} = -\dfrac{\partial \bar{B}}{\partial t} \\ \nabla \times \bar{B} = \mu_0 \bar{J} + \mu_0 \varepsilon_0 \dfrac{\partial \bar{E}}{\partial t} \end{cases}$$

(the right hand side of the Vlasov equation equal to zero means that we have no collisions)
We define

$$\rho_{m,\alpha} = m_\alpha n_\alpha; \qquad \rho_{c,\alpha} = q_\alpha n_\alpha; \qquad \bar{J} = \sum_\alpha q_\alpha m_\alpha \bar{U}_\alpha$$

Now we integrate in the velocities and we move to the two fluids description.

Zeroth moment equation

$$\int \left\{ \frac{\partial f_\alpha}{\partial t} + \bar{v} \cdot \nabla f_\alpha + \frac{q_\alpha}{m_\alpha}\left[\bar{E} + \bar{V} \times \bar{B}\right] \cdot \nabla_{\bar{v}} f_\alpha \right\} d^3v = 0$$

Wich we solve one integral at a time
The first one is

$$\int \frac{\partial f_\alpha}{\partial t}\, d^3v = \frac{\partial}{\partial t} \int f_\alpha\, d^3v = \frac{\partial n_\alpha(\bar{x}, t)}{\partial t}$$

The second integral is

$$\int \bar{v} \cdot \nabla f_\alpha\, d^3v = \nabla \cdot \left\{ \int \bar{v} f_\alpha\, d^3v \right\} = \nabla \cdot (n_\alpha \bar{U}_\alpha)$$

To solve the third integral we note that $\bar{E}$ and $\bar{B}$ are independent of the velocity.
So $\bar{V} \times \bar{B}$ is perpendicular to the velocity.
The third integral reduces to

$$\begin{aligned} &\int \frac{q_\alpha}{m_\alpha}\left[\bar{E} + \bar{V} \times \bar{B}\right] \cdot \nabla_{\bar{v}} f_\alpha\, d^3v = \\ &= \frac{q_\alpha}{m_\alpha} \int \nabla_{\bar{v}} \cdot \left\{ \left[\bar{E} + \bar{V} \times \bar{B}\right] f_\alpha \right\} d^3v = \\ &= \frac{q_\alpha}{m_\alpha}\left[\bar{E} + \bar{V} \times \bar{B}\right] f_\alpha \Big|_{v \to \pm\infty} = \\ &= 0 \end{aligned}$$

Because f_α goes to zero at infinity faster than the other terms.
So we get

$$\frac{\partial n_\alpha}{\partial t} + \nabla \cdot (n_\alpha \bar{U}_\alpha) = 0$$

We are facing now a **closure problem**, we have a moment which depends on the next moment.

First moment equation

The first moment equation will be given by

$$\int \left\{ \bar{v} \cdot \frac{\partial f_\alpha}{\partial t} + \bar{v} \cdot \nabla(f_\alpha \bar{v}) + \frac{q_\alpha}{m_\alpha} \bar{v} \cdot \nabla_{\bar{v}} \left[\left(\bar{E} + \bar{V} \times \bar{B} \right) f_\alpha \right] \right\} d^3v = 0$$

The first integral is

$$\int \bar{v} \cdot \frac{\partial f_\alpha}{\partial t} \, d^3v = \int \frac{\partial}{\partial t} (f_\alpha \bar{v}) \, d^3v = \frac{\partial}{\partial t} \left(n_\alpha \bar{U}_\alpha \right)$$

The second integral is

$$\int \bar{v} \cdot \nabla(f_\alpha \bar{v}) \, d^3v = \int \nabla \cdot (f_\alpha \bar{v}\bar{v}) \, d^3v$$

with

$$f_\alpha \bar{v}\bar{v} = \begin{pmatrix} f_\alpha v_x v_x & f_\alpha v_x v_y \\ f_\alpha v_y v_x & f_\alpha v_y v_y \end{pmatrix}$$

We now define the **pressure tensor**

$$\overset{\leftrightarrow}{P^\alpha} = m_\alpha \int \left(\bar{v} - \bar{U}_\alpha \right) \left(\bar{v} - \bar{U}_\alpha \right) f_\alpha \, d^3v =$$

$$= m_\alpha \int \left[\bar{v}\bar{v} - \bar{U}_\alpha \bar{v} - \bar{v}\bar{U}_\alpha + \bar{U}_\alpha \bar{U}_\alpha \right] f_\alpha \, d^3v =$$

$$= m_\alpha \int \bar{v}\bar{v} f_\alpha \, d^3v - m_\alpha n_\alpha \bar{U}_\alpha \bar{U}_\alpha \pm \cancel{m_\alpha n_\alpha \bar{U}_\alpha \bar{U}_\alpha} =$$

$$= m_\alpha \int \bar{v}\bar{v} f_\alpha \, d^3v - m_\alpha n_\alpha \bar{U}_\alpha \bar{U}_\alpha$$

So finally, the second integral is

$$\int \bar{v} \cdot \nabla(f_\alpha \bar{v}) \, d^3v = \int \nabla \cdot (f_\alpha \bar{v}\bar{v}) \, d^3v = \nabla \cdot \left[\frac{\overset{\leftrightarrow}{P^\alpha}}{m_\alpha} + n_\alpha \bar{U}_\alpha \bar{U}_\alpha \right]$$

For the third integral we have (solving by parts)

$$\int \frac{q_\alpha}{m_\alpha} \bar{v} \cdot \nabla_{\bar{v}} \left[\left(\bar{E} + \bar{V} \times \bar{B} \right) f_\alpha \right] d^3v =$$

$$= \frac{q_\alpha}{m_\alpha} \left\{ \bar{v} \left[\left(\bar{E} + \bar{V} \times \bar{B} \right) f_\alpha \right] \cdot \hat{n} \Big|_{-\infty}^{+\infty} - \int \left(\bar{E} + \bar{V} \times \bar{B} \right) f_\alpha \, d^3v \right\} =$$

$$= \frac{q_\alpha}{m_\alpha} n_\alpha \left[\bar{E} + \bar{U}_\alpha \times \bar{B} \right]$$

The final two fluid equations

So in the end we have

$$\frac{\partial}{\partial t}\left(n_\alpha \bar{U}_\alpha\right) + \nabla \cdot \left[\frac{\overleftrightarrow{P}^\alpha}{m_\alpha} + n_\alpha \bar{U}_\alpha \bar{U}_\alpha\right] + \frac{q_\alpha}{m_\alpha} n_\alpha \left[\bar{E} + \bar{U}_\alpha \times \bar{B}\right] = 0$$

with the pressure tensor being

$$\overleftrightarrow{P}^\alpha = P_{ij}^\alpha = \begin{pmatrix} P_{xx}^\alpha & P_{xy}^\alpha & \dots \\ P_{yx}^\alpha & P_{yy}^\alpha & \dots \\ \dots & \dots & \dots \end{pmatrix}$$

Wich is symmetric and diagonalizable.

"Pressure P and temperature T are the same"

It can be:

Isotropic

It has equal entries on the diagonal ($\overleftrightarrow{P}^\alpha = T\mathbb{1}$)

Spherical symmetry

Anisotropic

$T_{xx} > T_{yy}$

Agyrotropic

$T_{xx} \neq T_{yy} \neq T_{zz}$

It has the shape of a confetto, ellissoide with three different axis.

Real data from space gives all terms of the pressure tensor different from zero.

We will study the easiest case with $\overleftrightarrow{P}^\alpha = P_\alpha \mathbb{1}$.

The final system will be

$$\begin{cases} \dfrac{\partial n_\alpha}{\partial t} + \nabla \cdot (n_\alpha \bar{U}_\alpha) = 0 \\ \dfrac{\partial}{\partial t}\left(m_\alpha n_\alpha \bar{U}_\alpha\right) + \nabla \cdot \left[m_\alpha n_\alpha \bar{U}_\alpha \bar{U}_\alpha\right] + \nabla P_\alpha + q_\alpha n_\alpha \left[\bar{E} + \bar{U}_\alpha \times \bar{B}\right] = 0 \end{cases}$$

which we close with an Equation of State

$$P_\alpha = A n_\alpha^\gamma$$

plus the Maxwell equations.

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